RELATIVE PSEUDO-COHERENCE OF A COMPLEX ON A CLOSED SUBSCHEME OF A SMOOTH CHART EQUALS ABSOLUTE PSEUDO-COHERENCE OF THE PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17274-17292 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17294-17306, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. This is the chart-transfer principle that lets the whole relative pseudo-coherence theory be read inside smooth affine charts; it is the second use of L16115 and first load-bearing use of L16133, both on the shelf after this wave.

Allowed standard prerequisites (editorial).

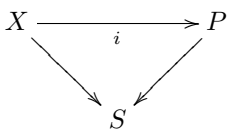
- Explicit prerequisite boundary. Chapter lemmas finite-morphism-relative-pseudo-coherence (L16133, same wave) and composition-relative-pseudo-coherent (L16115); More on Algebra, lemma flat-finite-presentation-perfect; the definition section of relative pseudo-coherence.

Source reference. Original lemma 09UT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17235–17245. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09UK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16995–17001. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 067J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17306–17306. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let*



Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

be a commutative diagram of schemes. Assume i is a closed immersion and $P \to S$ flat and locally of finite presentation. Let E be an object of $D(\mathcal{O}_X)$. Then the following are equivalent

(1) *E is m-pseudo-coherent relative to S,*
(2) *Ri_*E is m-pseudo-coherent relative to S, and*
(3) *Ri_*E is m-pseudo-coherent on P.*

Proof. The equivalence of (1) and (2) is Lemma 09UK. The equivalence of (2) and (3) follows from Lemma 09UT applied to $\mathrm{id} : P \to P$ provided we can show that $\mathcal{O}_P$ is pseudo-coherent relative to S. This follows from More on Algebra, Lemma 067J and the definitions. □